

Purva Chiniya

☎ (240) 501-3488 | 📄 Google Scholar | 🔗 LinkedIn | ✉ purvachiniya15@gmail.com | 📍 Bellevue, WA

EDUCATION

University of Maryland, College Park

Master of Science in Computer Science

Courses: Distributed Systems for ML, Advanced Numerical Optimization, LLM Security, ML at Scale

College Park, MD

Aug 2023 – May 2025

Indian Institute of Technology Roorkee

Bachelor of Technology in Electrical Engineering

Awards: Best Project Award, Top Innovation Award

Roorkee, India

Aug 2017 – June 2021

EXPERIENCE

Amazon.com

Applied Scientist Amazon Ads

Bellevue, WA

June 2025 – Present

- Architected a **multi-agent autonomous framework** for **Sponsored Products** using **LLM-based reasoning agents** to automate campaign optimizations for 1M+ advertisers; engineered scalable data pipelines in **PySpark** and **AWS Glue** to synthesize multi-terabyte seller and customer telemetry from **S3** data lakes.
- Deployed a runtime safety model for **Alexa+** using a **vLLM** inference engine handling **80M+ daily requests** at 35ms P90 latency; optimized throughput utilizing **continuous batching** and **PagedAttention**.
- Directed the alignment strategy using **Adversarial Fine-Tuning (AFT)** on 7b+ models, boosting precision **8x (4% → 34%)** and cutting False Positive Rate by **97%** to effectively mitigate reasoning over-refusals.

Sentient Foundation

AI Research Intern

Remote, US

Dec 2024 – May 2025

- Built **Open Deep Search**, a reasoning agent achieving **SOTA (75.3%)** on FRAMES and outperforming **GPT-4o Search Preview** via adaptive planning and dynamic query rewriting for complex multi-hop retrieval.
- Built the **CodeAct** (Chain-of-Code Action) orchestration layer using **SmolAgents** and LiteLLM to execute multi-step reasoning; implemented a reproducible evaluation harness comparing **ReAct** vs. **CodeAct**.
- Developed an **HPO** hyperparameter optimisation fine-tuning pipeline for data-mixture and hyperparameter search, cutting tuning iterations by **30%** and reducing fine-tuning cycle time by **40%** across **3B–70B** parameter models.

Amazon.com

Applied Scientist Intern

Bellevue, WA

May 2024 – Aug 2024

- Engineered a novel **inference-time safety steering** mechanism utilizing classifier gradients to guide decoding, achieving a **20%** reduction in Attack Success Rate (ASR) on AdvBench while cutting false refusal rates by **50%**.
- Benchmarked performance against SOTA defenses (SafeDecoding, DPO) on Llama2-7B, validating the method's effectiveness on jailbreak robustness; research accepted for publication at the **AMLC 2024** conference and **LREC 2026** conference.

Cisco Systems

Software Engineer II

Bangalore, India

Aug 2021 – Aug 2023

- Optimized **C++ BGP routing modules** for **Google peering** traffic, resolving critical concurrency bottlenecks to drive a **16% revenue increase** for the XR Business Unit via enhanced service availability.

RESEARCH PUBLICATIONS

- [1] **Gradient-Controlled Decoding: A Safety Guardrail for LLMs with Dual-Anchor Steering**. Accepted at **LREC'26**.
- [2] **SecRepoBench: Benchmarking Secure Code Generation in Real-World Repositories**. Repo-level vuln benchmark spanning **27 C/C++ repositories** (318 tasks); revealed a **31%** security drop vs single-file baselines. Accepted at **LLM4Code 2026**. arXiv:2504.21205.
- [3] **Open Deep Search: SOTA Agentic Search via Planning**. Achieved **75.3%** on **FRAMES** and outperformed **GPT-4o Search Preview** by **+9.7%** via adaptive planning and tool use. **Preprint (2025)**. arXiv:2503.20201.
- [4] **AV-RIR: Audio-Visual Room Impulse Response Estimation**. Multi-task learning fusing audio-visual cues to reconstruct acoustic environments. Accepted at **CVPR 2024**. arxiv:2312.00834.
- [5] **CoSyn: Implicit Hate Speech Detection via Hyperbolic Networks**. Hyperbolic GNN (GCN) + TreeLSTM to model latent social dynamics in conversational threads. Accepted at **EMNLP 2023**. arXiv:2303.03387.
- [6] **LipGER: Visually-Conditioned ASR Error Correction**. Visual modality pipeline to correct ASR errors in noise-heavy environments and improve robustness. Accepted at **INTERSPEECH 2024**. arXiv:2406.04432

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Bash

ML/Systems & Tools: PyTorch, Triton, vLLM, DeepSpeed, FlashAttention, Codex, Git, Docker, Cursor, Claude Code